

Harsh Dave

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Education

Texas A&M University

M.S. Computer Science

College Station, TX

Expected 2028

Texas A&M University

B.S. Computer Science & B.S. Statistics, Summa Cum Laude

College Station, TX

May 2026 | GPA: 3.92/4.00

Research Interests

Computational genomics of *Mycobacterium tuberculosis*; molecular evolution and positive selection; tuberculosis–diabetes comorbidity; drug resistance.

Research Experience

Graduate Assistant Research (Computational Genomics)

June 2026 – Present

Ioerger Lab, Texas A&M University (Advisor: Thomas R. Ioerger)

College Station, TX

- Fit gene-level linear models to TnSeq fitness under rifampicin and other growth conditions, compare nested models with likelihood-ratio tests, and apply false-discovery-rate correction. Tools: Python, pandas, SciPy, statsmodels.
- Analysis code and the programs it runs: pN/pS and codeml inputs, GenomeMap, PAML, TTN-Fitness, and TRANSIT.
- Investigating evidence of positive selection in *M. tuberculosis* genomes from diabetes and non-diabetes patient cohorts, analyzing approximately **4,000 genes** to identify candidate genomic differences.
- Built reproducible **Python and Slurm** pipelines for cohort QC, FASTA processing, Bayesian dN/dS estimation with **GenomeMap**, and posterior comparisons across large-scale TAMU HPRC runs.
- Validated selection signals with pN/pS enrichment, chi-square testing, Benjamini-Hochberg and local-FDR correction, drug-resistance positive controls, and independent **PAML/codeml** site-model analyses.

Undergraduate Researcher

January 2025 – May 2025

UrbanResilience.AI Lab, Texas A&M University

College Station, TX

- Built **Python** ingestion and preprocessing pipelines integrating air-quality sensor APIs and web-scraped wildfire data across Los Angeles sources for downstream emergency-response analysis.

Manuscripts in Preparation

Dave, H., Shatby, A., and Ioerger, T. Differential Bayesian analysis of genomic sequences identifies *M. tuberculosis* genes under positive selection in TB patients with diabetes.

Teaching Experience

Teaching Assistant, CS 111

Spring 2026

Texas A&M University

College Station, TX

- Supported students in **Java** and object-oriented programming through labs, debugging help, and weekly assignment review.
- Reviewed weekly submissions with consistent feedback and explained bugs through questions and small code examples.

Additional Experience

Software Engineering Capstone – Amazon-Sponsored

August 2025 – December 2025

Texas A&M University

College Station, TX

- Built a breach-intelligence analytics platform in a **3-person team**, implementing **Scrapy and Tor** ingestion plus a **LangChain and Gemini 2.5 Flash Lite** pipeline for entity extraction and threat classification.
- Designed storage and retrieval across **AWS S3, MongoDB, Elasticsearch, Redis, and Neptune**; delivered Streamlit and Kibana views for querying unstructured breach data by sector and geography.

Cloud Team Research Member

Early undergraduate

SpaceCraft VR

College Station, TX

- Helped an eight-person research team automate cloud infrastructure for simulation work.
- Contributed to a **React and TypeScript** sandbox and hardened access flows with **Auth0**.

AI Engineering Intern

May 2024 – August 2024

Videomagic

Remote

- Developed and evaluated **PyTorch and Hugging Face Transformers** computer-vision workflows for AI-generated video detection, analyzing frame-level outputs and model failure modes.
- Built **Feathers.js, MySQL, and Knex.js** APIs for model inference and data management, secured with **Auth0** and **JWT RBAC**.

Technical Skills

Data Science & Statistics: Python, R, SQL, pandas, NumPy, SciPy, scikit-learn, Bayesian inference, MCMC, hypothesis testing, FDR correction, Matplotlib
Bioinformatics & HPC: GenomeMap, PAML/codeml, dN/dS, pN/pS, FASTA pipelines, Slurm, Linux, TAMU HPRC
Machine Learning & Data Systems: PyTorch, Hugging Face Transformers, LangChain, AWS, MongoDB, Elasticsearch, Redis, Git, GitHub Actions
Programming: Java, C++, TypeScript, JavaScript, React, FastAPI, Docker, PostgreSQL
Languages: English, Hindi, Gujarati

Software

TB Genome Portal

orca2.tamu.edu/U19

H37Rv reference browser

- The lab catalog for annotations, operons, TnSeq, and GenomeMap plots across **4,018** genes.

MtbScope

harsh.bet/genes

H37Rv gene catalog

- Search **4,018** protein-coding genes and compare at most **8**. Annotations from TBDB, RefSeq, PATRIC, TubercuList, and NCBI. Literature from Europe PMC.
- Ranking is a weighted mean over the signals measured for that gene.

Research

harsh.bet/research

Open-access paper reader

- Fetch open-access full text from Europe PMC, then NCBI, and keep JATS tables, figures, and equations with the paper. A lookup list stops at **120** identifiers.
- Parse the first **80** PDF pages in the browser with PDF.js. Study decks live in Quizlet.

Quizlet

harsh.bet/quizlet

Notes, cards, quiz, blanks, match

- Notes, flashcards, quiz, blanks, and matching. One deck has **140** cards across terms, rules, theorems, and examples. Another has **41** statistics cards.

Simplfy

harsh.bet/simplfy

Statistics and tuberculosis lessons

- **288** plates: **168** tuberculosis and **120** statistics. A catalogue lesson is teach, example, practice, say-it-back, shelf, papers. Each plate has three checks.
- Each plate has an analogy, one worked example, and three checks.

Radar

harsh.bet/radar

Static snapshots, rebuilt twice a day

- Papers from Europe PMC, PubMed, bioRxiv, medRxiv, OpenAlex, and arXiv. Events from the Texas A&M calendar. Studies from Aggie Research Volunteers and ClinicalTrials.gov.
- Duplicates merge on identity keys, then on title. Studies rank by guaranteed dollars per hour from completion and attendance pay.

PickLedger

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Public JSON board

- **8** team-model and **3** player-prop pipelines write an immutable pregame ledger. ESPN grading runs every **15 minutes**. Scores use Brier, log loss, and ECE.
- Scores use the price recorded before the game, with Brier, log loss, and ECE.

Slate, Daymark, and Notes

Tasks, habits, notes

Local-first tasks, habits, and notes

- **Slate** is a task board with a 25/5/15 focus timer. **Daymark** tracks checks, counts, duration, quantity, or distance, with streaks that follow the target period.
- **Notes** is rich text with folders, labels, search, and a trash that can be restored.

Fare and Recipes

Meals

Nutrition recorded on the log line

- **Fare** stores the nutrition on each log line. Search includes **5,403** USDA foods. Barcodes use Open Food Facts.
- **Recipes** is **100** one-serving ovo-lacto meals at **800–1,300** kcal. Totals come from per-100 g ingredient rows, with grocery notes for each ingredient.

Gym, ShotLab, Timer, and Degree

Training and planning

Training, form, focus, and degree planning

- **Gym** keeps the exercise list on the logged day when the weekly template changes. A personal record is the highest set volume. The movement list has **876** records.
- **ShotLab** measures form on the phone with MediaPipe pose, 33 landmarks, and reports drift in shoulder widths. **Timer** draws eight canvas scenes and stores a wall-clock end time.
- **Degree** checks the MSCS thesis rules: 30 countable credits, 18 graded CSCE, at most 6 credits of CSCE 691, and three breadth areas. Catalog dated July 28, 2026.

Competitions and Course Projects

Alpha, Tidal Hackathon

1st place

React, TypeScript, multi-model tools

- Math-learning app in React and TypeScript, calling Gemini, OpenRouter, Claude, Manim, Desmos, and Wolfram Alpha. Reported tool-call failures fell from **25% to 6%** after output normalization and validation.

TAMU Datathon

2nd of 50 teams

Web scraping and AI tools

2024

- Used web scraping and AI tools to investigate clues during the competition.
- Placed **2nd of 50 teams** in a prompt-engineering challenge.

Sign Sense

Hackathon

YOLOv5, SageMaker, Svelte, FastAPI

- A sign-language practice app. The team trained a YOLOv5 model on SageMaker and served recognition through FastAPI to a Svelte interface.
- Connected the model to a camera-based practice flow so a learner gets feedback during a lesson.

ProfFinder, HowdyHack

24-hour build

SQL, JavaScript

- Faculty and course search over a database assembled from Texas A&M data, with schedule cards and section search.
- Assembled the course database during the hackathon and used it for class, section, and schedule search.

Point of Sale System

Scrum lead

React, PostgreSQL, AWS, OAuth2

- Led delivery of a point-of-sale platform covering ordering, inventory, analytics, authentication, and accessible customer flows.
- Integrated **OAuth2** and kept customer flows aligned with **WCAG 2.1**.

Coursework

CSCE 671, Computer-Human Interaction

Fall 2026

Texas A&M University

Graduate

- How people use software, how interfaces are designed, and how to evaluate whether a design works.
- Topics include accessibility, usability evaluation, and reading HCI research.

CSCE 627, Theory of Computability

Fall 2026

Texas A&M University

Graduate

- Automata, Turing machines, decidability, and reductions.
- Topics include finite automata, formal languages, and computational complexity.